CSS2112 WEB APPLICATION DEVELOPMENT

FINAL EXAMINATION

4th December 2024

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer Three (3) questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

Web applications are an essential part of modern-day business activities that help provide goods and services to consumers while providing revenue to these service providers. It can't be overemphasized that web applications such as Amazon, eBay, and Alibaba provide E-commerce platforms that allow consumers from all over the world to purchase goods and services and have those services or goods delivered to them. E-commerce web applications provide convenient means of shopping online. Users or consumers can browse through various items or services provided by that e-commerce website and if satisfied the consumers can add the item to the cart. Thereafter, the consumers can proceed to make a payment that is safe and secure using online payment processes as defined by the application stakeholders.

a) Describe the web development process framework with the aid of a diagram (10 marks)

b) Describe the 5 steps used in accessing the security of web Applications (10 marks)

c) Explain how JSON works concerning web applications (5 marks)

d) What are the two main technologies that define the World Wide Web? (2 marks)

e) Explain how the WWW works. (5 marks)

f) What are the four technologies that AJAX uses? Explain how they work in sites that are AJAX enabled (8 marks)

(Total: 40 marks)

SECTION B: Attempt any THREE questions in this section

Question 2

Zambia is among many of the third-world developing countries that are prominently dependent on the agriculture sector for food and revenue generation which comprises crops. Currently, the challenge is finding reliable consumers of crops that farmers produce. A web app has been suggested and you are asked to develop a prototype.

a) Explain the analysis of requirements of the system that you would gather (6 marks)
b) Draw a use case diagram for the web app ***(based on your analysis in question a) (9 marks)***
c) Write the code you would use to validate identification and authorization (5marks)

(20 marks)

Question 3

We are in the new technology era. So Mobile browsing is rocking. This means web pages on mobile browsers must be tested. Compatibility issues may be there with mobile devices as well.

a) Explain the term **responsive design** when it comes to websites. (3 marks)
b) Describe Compatibility, Security, and Performance testing concerning websites (9 marks)
c) What are the benefits of AJAX? (8 marks)

(20 marks)

Question 4

Any code that can respond to HTTP requests can run on a server. Here are a few other attributes of server-side code: Is never seen by the user (except within a rare malfunction), Stores data such as user profiles, tweets, pages, etc… and creates the page the user requested.

a) Explain the best practices for good web application architecture (6 marks)
b) Write an example of your choice using HTML 5 showing a basic HTML page (6 marks)
c) What are the similarities between JSON and XML (4 marks)
d) State the difference between JSON and XML (3 marks)
e) Define the term "HTTP". (1 mark)

(20 marks)

Question 5

Securing web applications is done at all the stages involved in the development of web applications. The design stage involves not only the design of the application under the specifications outlined in the first stage, but also defining secure coding standards, performing threat modeling, and developing a security architecture for the application.

a) Discuss how one can secure the coding standards at the design stage (**THREAT MODELING**)
(5 marks)

b) Briefly explain how the web application architecture works with the aid of a diagram
(10 marks)

c) State any **5** attributes of web applications (5 marks)

(20 marks)

END OF EXAMINATION